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Chapter 2

Estimation

2.1 Linear Model

Let's start by defining what is meant by a linear model. Suppose we want to model the response Y in terms of three predictors, X_1, X_2 and X_3. One very general form for the model would be:

$$Y=f(X1,X2,X3)+\varepsilon$$

where f is some unknown function and e is the error in this representation. ε is additive in this instance, but could enter in some even more general form. Still, if we assume that f is a smooth, continuous function, that still leaves a very wide range of possibilities. Even with just three predictors, we typically will not have enough data to try to estimate f directly. So we usually have to assume that it has some more restricted form, perhaps linear as in:

$$Y=\beta 0+\beta 1X1+\beta 2X2+\beta 3X3+\varepsilon$$

where β_i, i = 0,1,2,3 are unknown *parameters.* Unfortunately this term is subject to some confusion as engineers often use the term *parameter* for what statisticians call the variables, Y, X_1 and so on. β_0 is called the *intercept* term.

Thus the problem is reduced to the estimation of four parameters rather than the infinite dimensional f. In a linear model the *parameters enter linearly* — the predictors themselves do not have to be linear. For example:

$$Y=\beta 0+\beta 1X1+\beta 2\log X2+\beta 3X1X2+\varepsilon$$

is a linear model, but:

$$Y=\beta 0+\beta 1X1\beta 2+\varepsilon$$

is not. Some relationships can be transformed to linearity — for example, y=β0x1βε can be linearized by taking logs. Linear models seem rather restrictive, but because the predictors can be transformed and combined in any way, they are actually very flexible. The term *linear* is often used in everyday speech as almost a synonym for simplicity. This gives the casual observer the impression that linear models can only handle small simple datasets. This is far from the truth — linear models can easily be expanded and modified to handle complex datasets. *Linear* is also used to refer to straight lines, but linear models can be curved. Truly nonlinear models are rarely absolutely necessary and most often arise from a theory about the relationships between the variables, rather than an empirical investigation.

Where do models come from? We distinguish several different sources:

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1. Physical theory may suggest a model. For example, Hooke's law says that the extension of a spring is proportional to the weight attached. Models like these usually arise in the physical sciences and engineering.
2. Experience with past data. Similar data used in the past were modeled in a particular way. It is natural to see whether the same model will work with the current data. Models like these usually arise in the social sciences.
3. No prior idea exists — the model comes from an exploration of the data. We use skill and judgment to pick a model. Sometimes it does not work and we have to try again.

Models that derive directly from physical theory are relatively uncommon so that usually the linear model can only be regarded as an approximation to a complex reality. We hope it predicts well or explains relationships usefully but usually we do not believe it is exactly true. A good model is like a map that guides us to our destination. For the rest of this chapter, we will stay in the special world of Mathematics where all models are true.

2.2 Matrix Representation

We want a general solution to estimating the parameters of a linear model. We can find simple formulae for some special cases but to devise a method that will work in all cases, we need to use matrix algebra. Let's see how this can be done.

We start with some data where we have a response Y and, say, three predictors, X_1, X_2 and X_3. The data might be presented in tabular form like this:

y1x11x12x13y2x21x22x23……ynxn1xn2xn3

where n is the number of observations, or *cases*, in the dataset.

Given the actual data values, we may write the model as:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \varepsilon_i \quad i = 1, \ldots, n.$$

It will be more convenient to put this in a matrix/vector representation. The regression equation is then written as:

y=Xβ+ε

where $y = (y_1 \ldots, y_n)^T$, $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_n)^T$,=$\beta = (\beta_0, \ldots, \beta_3)^T$ and:

X=(1x11x12x131x21x22x23……1xn1xn2xn3)

The column of ones incorporates the intercept term. One simple example is the *null model* where there is no predictor and just a mean $y = \mu + \varepsilon$:

(y1…yn)=(1…1)μ+(ε1…εn)

We can assume that $E\varepsilon = 0$ since if this were not so, we could simply absorb the nonzero expectation for the error into the mean to get a zero expectation.

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2.3 Estimating β

The regression model, y = *X*β + ε, partitions the response into a systematic component *X*β and a random component ε. We would like to choose p so that the systematic part explains as much of the response as possible. Geometrically speaking, the response lies in an n-dimensional space, that is, y # I$\mathbf{R}^n$ while p G I$\mathbf{R}^p$ where *p* is the number of parameters. If we include the intercept then *p* is the number of predictors plus one. It is easy to get confused as to whether *p* is the number of predictors or parameters, as different authors use different conventions, so be careful.

The problem is to find β so that *X*β is as close to *Y* as possible. The best choice, the estimate β^ apparent in the geometrical representation seen in Figure 2.1. β^ is, in this sense, the best estimate of β within the model space. The β^ values are sometimes called the regression coefficients. The response predicted by the model is y^=Xβ^ or *Hy* where *H* is an orthogonal projection matrix. The y^ are called *predicted* or *fitted* values. The difference between the actual response and the predicted response is denoted by ε^ and is called the *residual*..

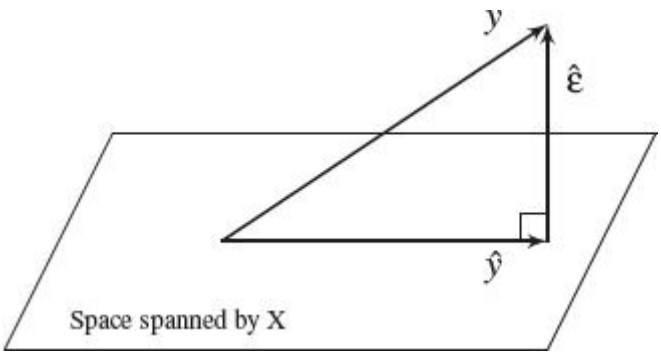

Figure 2.1 Geometrical representation of the estimation β. The data vector Y is projected orthogonally onto the model space spanned by *X*. The fit is represented by projection y^=Xβ^ with the difference between the fit and the data represented by the residual vector ε^.

The conceptual purpose of the model is to represent, as accurately as possible, something complex, *y*, which is n-dimensional, in terms of something much simpler, the model, which is *p*-dimensional. Thus if our model is successful, the structure in the data should be captured in those *p* dimensions, leaving just random variation in the residuals which lie in an (n − *p)*-dimensional space. We have:

Data=Systematic Structure + Random Variationn dimensions= pdimensions + (n−p) dimensions

2.4 Least Squares Estimation

The estimation of β can also be considered from a nongeometric point of view. We define the best estimate of β as the one which minimizes the sum of the squared errors:

∑εi2=εTε=(y−Xβ)T(y−Xβ)

Differentiating with respect to β and setting to zero, we find that β^ satisfies:

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XTXβ^=XTy

These are called the *normal equations.* We can derive the same result using the geometric approach. Now provided X^TX is invertible:

β^ =(XTX)−1XTyXβ^ =X(XTX)−1XTyy^=Hy

$H = X(X^TX)^{-1}X^T$ is called the *hat matrix* and is the orthogonal projection of y onto the space spanned by X. H is useful for theoretical manipulations, but you usually do not want to compute it explicitly, as it is an n × n matrix which could be uncomfortably large for some datasets. The following useful quantities can now be represented using H.

The predicted or fitted values are y^=Hy=Xβ^ while the residuals are ε^=y−xβ^=y−y^=(I−H)y. The residual sum of squares (RSS) is ε^Tε^=yT(I−H)(I−H)y=yT(I−H)y

Later, we will show that the least squares estimate is the best possible estimate of p when the errors ε are uncorrelated and have equal variance, i.e., var $\varepsilon = \sigma^2 I$. β^ is unbiased and has variance $(X^TX)^{-1}\sigma^2$ provided var $\varepsilon = \sigma^2 I$. Since β^ is a vector, its variance is a matrix.

We also need to estimate σ^2. We find that Eε^Tε^=σ2(n−p), which suggests the estimator:

σ^2=ε^Tε^n−p=RSSn−p

as an unbiased estimate of σ^2. $n - p$ is called the degrees of freedom of the model. Sometimes you need the standard error for a particular component of β^ which can be picked out as *se* (β^i−1)=(XTX)ii−1σ^.

2.5 Examples of Calculating β^

In a few simple models, it is possible to derive explicit formulae for p:

1. When $y = \mu + \varepsilon$, $X = \mathbf{1}$ and $\beta = \mu$ hence $X^TX = \mathbf{1}^T\mathbf{1} = n$ so:

β^=(XTX)−1XTy=1n1Ty=y¯

2. Simple linear regression (one predictor):

yi=β0+β1xi+εi(y1#yn)=(1x1#1xn)(β0β1)+(ε1#εn)

We can now apply the formula but a simpler approach is to rewrite the equation as:

yi=β0+β1x¯⌢β0′+β1(xi−x¯)+εi

so now:

X=(1x1−x¯#xn−x¯1) XTX=(n00∑i=1n(xi−x¯)2)

Next work through the rest of the calculation to reconstruct the familiar estimate, that is:

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$$\hat{\beta}_1 = \sum(x_i - \bar{x})y_i \sum(x_i - \bar{x})^2$$

In higher dimensions, it is usually not possible to find such explicit formulae for the parameter estimates unless X^TX happens to be a simple form. So typically we need computers to fit such models. Regression has a long history, so in the time before computers became readily available, fitting even quite simple models was a tedious time consuming task. When computing was expensive, data analysis was limited. It was designed to keep calculations to a minimum and restrict the number of plots. This mindset remained in statistical practice for some time even after computing became widely and cheaply available. Now it is a simple matter to fit a multitude of models and make more plots than one could reasonably study. The challenge now for the analyst is to choose among these intelligently to extract the crucial information in the data.

2.6 Example

Now let's look at an example concerning the number of species found on the various Galapagos Islands. There are 30 cases (Islands) and seven variables in the dataset. We start by reading the data into R and examining it:

```
> data(gala, package="faraway")
> head(gala[,-2])
             Species  Area Elevation Nearest Scruz Adjacent
Baltra            58 25.09       346     0.6   0.6     1.84
Bartolome         31  1.24       109     0.6  26.3   572.33
Caldwell           3  0.21       114     2.8  58.7     0.78
Champion          25  0.10        46     1.9  47.4     0.18
Coamano            2  0.05        77     1.9   1.9   903.82
Daphne.Major      18  0.34       119     8.0   8.0     1.84
```

The variables are Species — the number of species found on the island, Area — the area of the island (km^2), Elevation — the highest elevation of the island (m), Nearest — the distance from the nearest island (km), Scruz — the distance from Santa Cruz Island (km), Adjacent — the area of the adjacent island (km^2). We have omitted the second column (which has the number of endemic species) because we shall not use this alternative response variable in this analysis.

The data were presented by Johnson and Raven (1973) and also appear in Weis-berg (1985). I have filled in some missing values for simplicity (see Chapter 13 for how this can be done). Fitting a linear model in R is done using the lm() command. Notice the syntax for specifying the predictors in the model. This is part of the *Wilkinson-Rogers* notation. In this case, since all the variables are in the gala data frame, we must use the data= argument:

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```
> lmod <- lm(Species ~ Area + Elevation + Nearest + Scruz  + Adjacent,
     data=gala)
> summary(lmod)

Call:
lm(formula = Species ~ Area + Elevation + Nearest + Scruz +
    Adjacent, data = gala)

Residuals:
    Min      1Q  Median      3Q     Max
-111.68  -34.90   -7.86   33.46  182.58

Coefficients:
            Estimate Std. Error t value  Pr(>|t|)
(Intercept)  7.06822   19.15420    0.37    0.7154
Area        -0.02394    0.02242   -1.07    0.2963
Elevation    0.31946    0.05366    5.95 0.0000038 ***
Nearest      0.00914    1.05414    0.01    0.9932
Scruz       -0.24052    0.21540   -1.12    0.2752
Adjacent    -0.07480    0.01770   -4.23    0.0003 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 61 on 24 degrees of freedom
Multiple R-squared:  0.766,     Adjusted R-squared:  0.717
F-statistic: 15.7 on 5 and 24 DF,  p-value: 6.84e-07
```

For my tastes, this output contains rather too much information. I have written an alternative called sumary which produces a shorter version of this. Since we will be looking at lot of regression output, the use of this version makes this book several pages shorter. Of course, if you prefer the above, feel free to add the extra "m" in the function call. You will need to install and load my package if you want to use my version:

```
> require(faraway)
> sumary(lmod)
            Estimate Std. Error t value  Pr(>|t|)
(Intercept)  7.06822   19.15420    0.37    0.7154
Area        -0.02394    0.02242   -1.07    0.2963
Elevation    0.31946    0.05366    5.95 0.0000038
Nearest      0.00914    1.05414    0.01    0.9932
Scruz       -0.24052    0.21540   -1.12    0.2752
Adjacent    -0.07480    0.01770   -4.23    0.0003

n = 30, p = 6, Residual SE = 60.975, R-Squared = 0.77
```

We can identify several useful quantities in this output. Other statistical packages tend to produce output quite similar to this. One useful feature of R is that it is possible to directly calculate quantities of interest. Of course, it is not necessary here because the lm() function does the job, but it is very useful when the statistic you want is not part of the prepackaged functions. First, we extract the X-matrix:

```
> x <- model.matrix( ~ Area + Elevation + Nearest + Scruz  + Adjacent,
    gala)
```

and here is the response y:

```
> y <- gala$Species
```

Now let's construct $(X^TX)^{-1}$.t() does transpose and %*% does matrix multiplication. solve(A) computes A^{-1} while solve(A,b) solves $Ax = b$:

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```
> xtxi <- solve(t(x) %*% x)
```

We can get β^ directly, using $(X^TX)^{-1}X^Ty$:

```
> xtxi %*% t(x) %*% y
                  [,1]
1           7.068221
Area       -0.023938
Elevation   0.319465
Nearest     0.009144
Scruz      -0.240524
Adjacent   -0.074805
```

This is a very bad way to compute β^ It is inefficient and can be very inaccurate when the predictors are strongly correlated. Such problems are exacerbated by large datasets. A better, but not perfect, way is:

```
> solve(crossprod(x,x),crossprod(x,y))
                [,1]
1           7.068221
Area       -0.023938
Elevation   0.319465
Nearest     0.009144
Scruz      -0.240524
Adjacent   -0.074805
```

where crossprod(x,y) computes x^Ty. Here we get the same result as lm() because the data are well-behaved. In the long run, you are advised to use carefully programmed code such as found in lm() which uses the QR decomposition. To see more details, consult a text such as Thisted (1988) or read Section 2.7.

We can extract the regression quantities we need from the model object. Commonly used are residuals(), fitted(), df.residual() which gives the degrees of freedom, deviance() which gives the RSS and coef() which gives the cTβ^. You can also extract other needed quantities by examining the model object and its summary:

```
> names(lmod)
 [1] "coefficients"  "residuals"     "effects"       "rank"
 [5] "fitted.values" "assign"        "qr"            "df.residual"
 [9] "xlevels"       "call"          "terms"         "model"
> lmodsum <- summary(lmod)
> names(lmodsum)
 [1] "call"          "terms"         "residuals"     "coefficients"
 [5] "aliased"       "sigma"         "df"            "r.squared"
 [9] "adj.r.squared" "fstatistic"    "cov.unscaled"
```

We can estimate σ using the formula in the text above or extract it from the summary object:

```
> sqrt(deviance(lmod)/df.residual(lmod))
[1] 60.975
> lmodsum$sigma
[1] 60.975
```

We can also extract $(X^TX)^{-1}$ and use it to compute the standard errors for the coefficients. (diag() returns the diagonal of a matrix):

```
> xtxi <- lmodsum$cov.unscaled
> sqrt(diag(xtxi))*60.975
(Intercept)        Area   Elevation     Nearest       Scruz
  19.154139    0.022422    0.053663    1.054133    0.215402
   Adjacent
   0.017700
```

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or get them from the summary object:

```
>  lmodsum$coef[,2]
(Intercept)        Area   Elevation     Nearest       Scruz
  19.154198    0.022422    0.053663    1.054136    0.215402
   Adjacent
   0.017700
```

2.7 QR Decomposition

This section might be skipped unless you are interested in the actual calculation of β^ and related quantities. Any design matrix X can be written as:

$$X=Q(R0)=QfR$$

where Q is an $n \times n$ orthogonal matrix, that is $Q^TQ = QQ^T = I$ and R is a $p \times p$ upper triangular matrix ($R_ij = 0$ for $i > j$). The **0** is an $(n - p) \times p$ matrix of zeroes while Q_f is the first p columns of Q .

The $RSS = (y - X\beta)^T(y - X\mathbf{p}) = \|y - X\beta\|^2$ where || # || is the Euclidean length of a vector. The matrix Q represents a rotation and does not change length. Hence:

$$RSS=\|QTy-QTX\beta\|2=\|(fr)-(R0)\beta\|2$$

where (fr)=QTy for vector f of length p and vector r of length $n - p$. From this we see:

$$RSS=\|f-R\beta\|2+\|r\|2$$

which can be minimized by setting p so that $R\beta = f$.

Let's see how this works for the Galápagos data. First we compute the QR decomposition:

```
> qrx <- qr(x)
```

The components of the decomposition must be extracted by other functions. For example, we can extract the Q matrix using qr.Q:

```
> dim(qr.Q(qrx))
[1] 30  6
```

Notice that we do not need the whole n **x** n matrix for the computation. The first p columns suffice so qr.q returns what we call Q_f. We can compute f:

```
> (f <- t(qr.Q(qrx)) %*% y)
          [,1]
[1,] -466.8422
[2,]  381.4056
[3,]  256.2505
[4,]    5.4076
[5,] -119.4983
[6,]  257.6944
```

Solving $R\beta = f$ is easy because of the triangular form of R. We use the method of backsubstitution:

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```
> backsolve(qr.R(qrx),f)
           [,1]
[1,]  7.068221
[2,] -0.023938
[3,]  0.319465
[4,]  0.009144
[5,] -0.240524
[6,] -0.074805
```

where the results match those seen previously.

2.8 Gauss-Markov Theorem

β^ is a plausible estimator, but there are alternatives. Nonetheless, there are three good reasons to use least squares:

1. It results from an orthogonal projection onto the model space. It makes sense geometrically.
2. If the errors are independent and identically normally distributed, it is the maximum likelihood estimator. Loosely put, the maximum likelihood estimate is the value of β that maximizes the probability of the data that was observed.
3. The Gauss-Markov theorem states that p is the best linear unbiased estimate (BLUE).

To understand the Gauss-Markov theorem we first need to understand the concept of an *estimable function.* A linear combination of the parameters $\psi = c^T \beta$ is estimable if and only if there exists a linear combination $a^T y$ such that:

$$EaTy=cT\beta \ \forall\beta$$

Estimable functions include predictions of future observations, which explains why they are well worth considering. If X is of full rank, then all linear combinations are estimable.

Suppose $E\varepsilon = 0$ and var $\varepsilon = \sigma^2 I$. Suppose also that the structural part of the model, $EY = X\beta$ is correct. (Clearly these are big assumptions and so we will address the implications of this later.) Let $\psi = c^T \beta$ be an estimable function; then the Gauss–Markov theorem states that in the class of all unbiased linear estimates of ψ, ψ =cTβ^ has the minimum variance and is unique.

We prove this theorem. Suppose $a^T y$ is some unbiased estimate of $c^T \beta$ so that:

$$EaTy=cT\beta\forall\beta aTX\beta=cT\beta\forall\beta$$

which means that $a^T X = c^T$. This implies that c must be in the range space of X^T which in turn implies that c is also in the range space of $X^T X$ which means there exists a X such that $c = X^T X\lambda$ so:

$$cT\beta\hat{}=\lambda TXTX\beta\hat{}=\lambda TXTy$$

Now we can show that the least squares estimator has the minimum variance — pick an arbitrary estimate $a^T y$ and compute its variance:

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var(aTy)=var(aTy−cTβ^+cTβ^) =var(aTy−λTXTy+cTβ^)

=var(aTy−λTXTy)+var(cTβ^)+2cov(aTy−λTXTy,λTXTy)

but

cov(aTy−λTXTy,λTXTy)=(aT−λTXT)σ2IXλ =(aTX−λTXTX)σ2Iλ=(cT−cT)σ2Iλ=0

so

var(aTy)=var(aTy−λTXTy)+var(cTβ^)

Now since variances cannot be negative, we see that:

var(aTy)≥var(cTβ^)

In other words, cTβ^ has minimum variance. It now remains to show that it is unique. There will be equality in the above relationship if var $(a^T y - \lambda^T X^T y) = 0$ which would require that $a^T - \lambda^T X^T = 0$ which means that aTy=λTXTy=cTβ^. So equality occurs only if aTy=cTβ^ so the estimator is unique. This completes the proof.
The Gauss–Markov theorem shows that the least squares estimate cTβ^ is a good choice, but it does require that the errors are uncorrelated and have equal variance. Even if the errors behave, but are nonnormal, then nonlinear or biased estimates may work better. So this theorem does not tell one to use least squares all the time; it just strongly suggests it unless there is some strong reason to do otherwise. Situations where estimators other than ordinary least squares should be considered are:

1. When the errors are correlated or have unequal variance, generalized least squares should be used. See Section 8.1.
2. When the error distribution is long-tailed, then robust estimates might be used. Robust estimates are typically not linear in *y*. See Section 8.4.
3. When the predictors are highly correlated (collinear), then biased estimators such as ridge regression might be preferable. See Chapter11.

2.9 Goodness of Fit

It is useful to have some measure of how well the model fits the data. One common choice is R^2, the so-called *coefficient of determination* or *percentage of variance explained:*

R2=1−∑(y^i−yi)2∑(yi−y¯)2=1−RSSTotal SS(Corrected for Mean)

Its range is $0 \le R^2 \le 1$ — values closer to 1 indicating better fits. For simple linear regression $R^2 = r^2$ where *r* is the correlation between *x* and y. An equivalent definition is:

R2=∑(y^i−y¯)2∑(yi−y¯)2

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or

R2=cor2(y^,y)

The graphical intuition behind R^2 is seen in Figure 2.2. Suppose you want to predict y. If you do not know x, then your best prediction is #, but the variability in this prediction is high. If you do know *x*, then your prediction will be given by the regression fit. This prediction will be less variable provided there is some

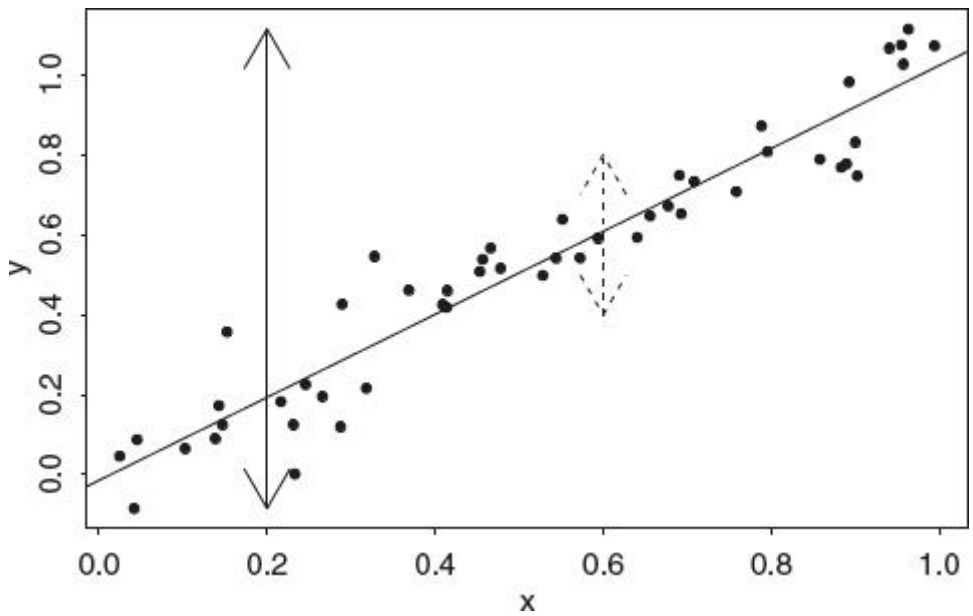

Figure 2.2 When *x* is not known, the best predictor of *y* is # and the variation is denoted by the dotted line. When *X* is known, we can predict *y* more accurately by the solid line. R^2 is related to the ratio of these two variances.

relationship between *x* and *y*. R^2 is one minus the ratio of the sum of squares for these two predictions. Thus for perfect predictions the ratio will be zero and R^2 will be one.

Some care is necessary if there is no intercept in your model. The denominator in the first definition of R^2 has a null model with an intercept in mind when the sum of squares is calculated. Unfortunately, R uses this definition and will give a misleadingly high R^2. If you must have an R^2 use the cor^2 (y, ^y) definition when there is no intercept.

What is a good value of R^2 ? It depends on the area of application. In the biological and social sciences, variables tend to be more weakly correlated and there is a lot of noise. We would expect lower values for R^2 in these areas — a value of, say, 0.6 might be considered good. In physics and engineering, where most data come from closely controlled experiments, we typically expect to get much higher R^2s and a value of 0.6 would be considered low. Some experience with the particular area is necessary for you to judge your R^2s well.

It is a mistake to rely on R^2 as a sole measure of fit. In Figure 2.3, we see some simulated datasets where the R^2 is around 0.65 for a linear fit in all four cases. The first plot on the upper left

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shows an ordinary relationship while on the right the variation in x is smaller as is the residual variation. However, predictions (within the range of *x*) would have less variation in the second case. On the lower left, the fit looks good except for two outliers demonstrating how sensitive R^2 is to a few extreme values. On the lower right, we see that the true relationship is somewhat quadratic which shows us that R^2 doesn't tell us much about whether we have the right model.

An alternative measure of fit is σ^. This quantity is directly related to the standard errors of estimates of β and predictions. The advantage is that σ^ is measured in the units of the response and so may be directly interpreted in the context of the particular dataset. This may also be a disadvantage in that one must understand the practical significance of this measure whereas R^2, being unitless, is easy to understand. The R regression summary returns both values and it is worth paying attention to both of them.

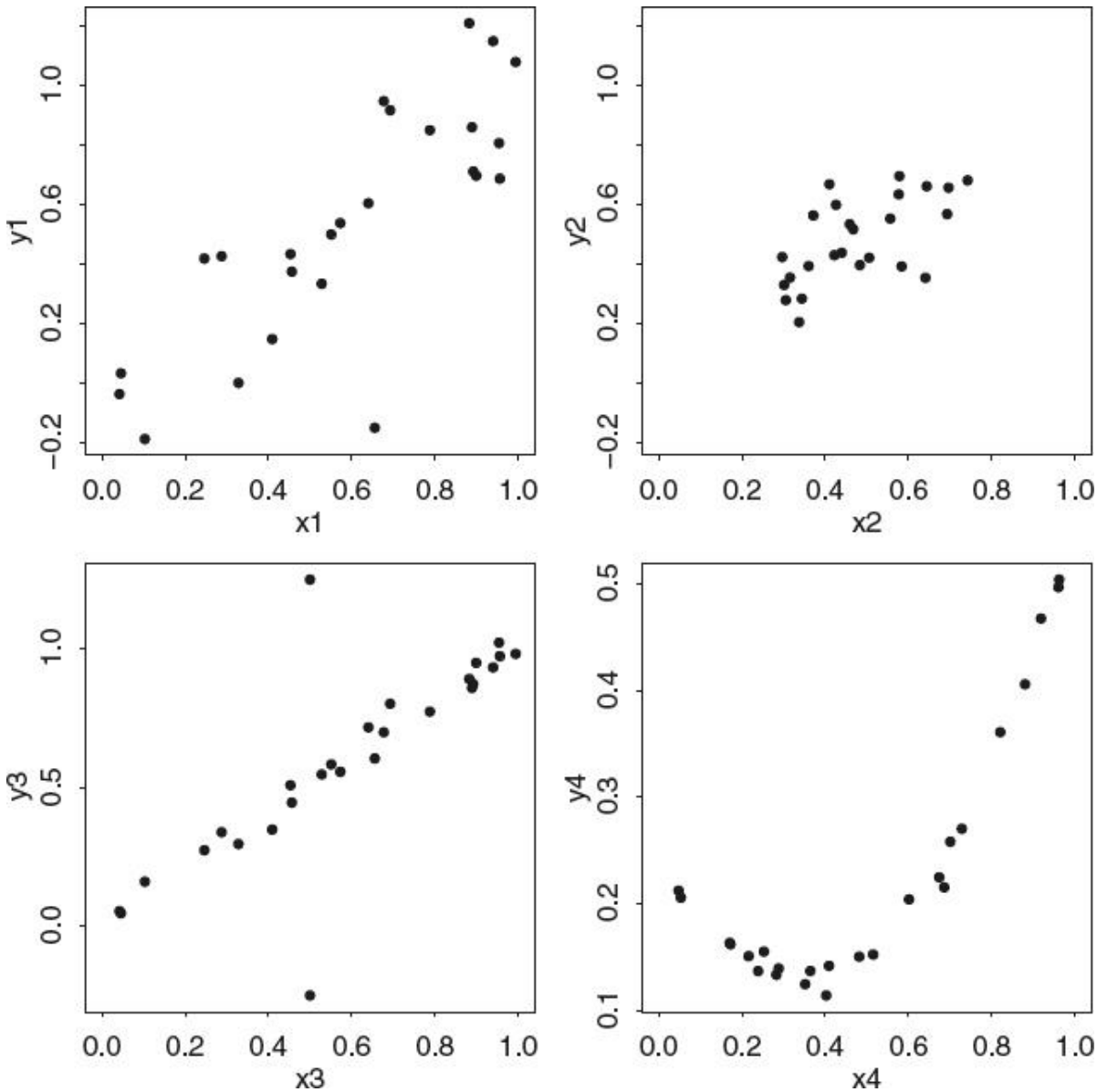

Figure 2.3 Four simulated datasets where R^2 is about 0.65. The plot on the upper left is well-behaved for R^2. In the plot on the upper right, the residual variation is smaller than the first plot but the variation in *x* is also smaller so R^2 is about the same. In the plot on the lower left, the fit looks strong except for a couple of outliers while on the lower right, the relationship is quadratic.

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2.10 Identifiability

The least squares estimate is the solution to the normal equations:

$$XTX\beta\hat{}=XTy$$

where X is an $n \times p$ matrix. If X^TX is singular and cannot be inverted, then there will be infinitely many solutions to the normal equations and β^ is at least partially unidentifiable. Unidentifiability will occur when X is not of full rank — that is when its columns are linearly dependent. With observational data, unidentifiability is usually caused by some oversight. Here are some examples:

1. A person's weight is measured both in pounds and kilos and both variables are entered into the model. One variable is just a multiple of the other.
2. For each individual we record the number of years of preuniversity education, the number of years of university education and also the total number of years of education and put all three variables into the model. There is an exact linear relation among the variables.
3. We have more variables than cases, that is, $p > n$. When p = n, we may perhaps estimate all the parameters, but with no degrees of freedom left to estimate any standard errors or do any testing. Such a model is called *saturated.* When $p > n$, then the model is sometimes called *supersaturated.* Such models are considered in large-scale screening experiments used in product design and manufacture and in bioinformatics where there are more genes than individuals tested, but there is no hope of uniquely estimating all the parameters in such a model. Different approaches are necessary.

Such problems can be avoided by paying attention. Identifiability is more of an issue in designed experiments. Consider a simple two-sample experiment, where the treatment observations are y_1, …, y_n and the controls are y_{n+1},…, y_{m+n}. Suppose we try to model the response by an overall mean μ and group effects α_1 and α_2:

$$yj=\mu+\alpha i+\varepsilon j \quad i=1,2 \quad j=1,\ldots,m+n(y1\ldots ynyn+1\ldots ym+n)=(110\ldots 110101...101)(\mu\alpha 1\alpha 2)+(\varepsilon 1\ldots\ldots\ldots\varepsilon m+n)$$

Now although X has three columns, it has only rank two — (μ, α_1, α_2) are not identifiable and the normal equations have infinitely many solutions. We can solve this problem by imposing some constraints, μ = 0 or $\alpha_1 + \alpha_2 = 0$, for example.

Statistics packages handle nonidentifiability differently. In the regression case above, some may return error messages and some may fit models because rounding error may remove the exact identifiability. In other cases, constraints may be applied but these may be different from what you expect. By default, R fits the largest identifiable model by removing variables in the reverse order of appearance in the model formula.

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Here is an example. Suppose we create a new variable for the Galapagos dataset — the difference in area between the island and its nearest neighbor:

```
> gala$Adiff <- gala$Area -gala$Adjacent
```

and add that to the model:

```
> lmod <- lm(Species ~ Area+Elevation+Nearest+Scruz+Adjacent +Adiff,
    gala)
> sumary(lmod)

Coefficients: (1 not defined because of singularities)
            Estimate Std. Error t value  Pr(>|t|)
(Intercept)  7.06822   19.15420    0.37    0.7154
Area        -0.02394    0.02242   -1.07    0.2963
Elevation    0.31946    0.05366    5.95 0.0000038
Nearest      0.00914    1.05414    0.01    0.9932
Scruz       -0.24052    0.21540   -1.12    0.2752
Adjacent    -0.07480    0.01770   -4.23    0.0003

n = 30, p = 6, Residual SE = 60.975, R-Squared = 0.77
```

We get a message about a singularity because the rank of the design matrix X is six, which is less than its seven columns. In most cases, the cause of identifia-bility can be revealed with some thought about the variables, but, failing that, an eigendecomposition of X^TX will reveal the linear combination(s) that gave rise to the unidentifiability — see Section 11.1.

Lack of identifiability is obviously a problem, but it is usually easy to identify and work around. More problematic are cases where we are close to unidentifiability. To demonstrate this, suppose we add a small random perturbation to the third decimal place of Adiff by adding a random variate from $U[-0.005,0.005]$ where U denotes the uniform distribution. Random numbers are by nature random so results are not exactly reproducible. However, you can make the numbers come out the same every time by setting the seed on the random number generator using set.seed(). I have done this here so you will not wonder why your answers are not exactly the same as mine, but it is not strictly necessary.

```
> set.seed(123)
> Adiffe <- gala$Adiff+0.001*(runif(30)-0.5)
```

and now refit the model:

```
> lmod <- lm(Species ~ Area+Elevation+Nearest+Scruz +Adjacent+Adiffe,
    gala)
> sumary(lmod)
               Estimate  Std. Error t value  Pr(>|t|)
(Intercept)      3.2964     19.4341    0.17      0.87
Area        -45122.9865  42583.3393   -1.06      0.30
Elevation        0.3130      0.0539    5.81 0.0000064
Nearest          0.3827      1.1090    0.35      0.73
Scruz           -0.2620      0.2158   -1.21      0.24
Adjacent     45122.8891  42583.3406    1.06      0.30
Adiffe       45122.9613  42583.3381    1.06      0.30

n = 30, p = 7, Residual SE = 60.820, R-Squared = 0.78
```

Notice that now all parameters are estimated, but the standard errors are very large because we cannot estimate them in a stable way. We set up this problem so we know the cause but in general we need to be able to identify such situations. We do this in Section 7.3.

2.11 Orthogonality

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Orthogonality is a useful property because it allows us to more easily interpret the effect of one predictor without regard to another. Suppose we can partition X in two, $X = [X_1 | X_2]$ such that X1TX2=0. So now:

$$Y=X\beta+\varepsilon=X1\beta1+X2\beta2+\varepsilon$$

and

$$XTX=(X1TX1X1TX2X2TX1X2TX2)=(X1TX100X2TX2)$$

which means:

$$\beta\hat{}1=(X1TX1)-1X1Ty\ \beta\hat{}2=(X2TX2)-1X2Ty$$

Notice that β^1 will be the same regardless of whether X_2 is in the model or not (and vice versa). So we can interpret the effect of X_1 without a concern for X_2. Unfortunately, the decoupling is not perfect. Suppose we wish to test $H_0 : \beta_1 = 0$. We have RSS/df=σ^2 that will be different depending on whether X_2 is included in the model or not, but the difference in F is not liable to be as large as in nonorthogonal cases.

If the covariance between vectors x_1 and x_2 is zero, then ∑j(xj1−x¯1)(xj2−x¯2)=. This means that if we center the predictors, a covariance of zero implies orthogonality. As can be seen in the second example in Section 2.5, we can center the predictors without essentially changing the model provided we have an intercept term.

Orthogonality is a desirable property, but will only occur when *X* is chosen by the experimenter. It is a feature of a good design. In observational data, we do not have direct control over *X* and this is the source of many of the interpretational difficulties associated with nonexperimental data.

Here is an example of an experiment to determine the effects of column temperature, gas/liquid ratio and packing height in reducing the unpleasant odor of a chemical product that was sold for household use. Read the data in and display:

```
> data(odor, package="faraway")
> odor
   odor temp gas pack
 1   66   -1  -1    0
 2   39    1  -1    0
 3   43   -1   1    0
 4   49    1   1    0
 5   58   -1   0   -1
 6   17    1   0   -1
 7   -5   -1   0    1
 8  -40    1   0    1
 9   65    0  -1   -1
10    7    0   1   -1
11   43    0  -1    1
12  -22    0   1    1
13  -31    0   0    0
14  -35    0   0    0
15  -26    0   0    0
```

The three predictors have been transformed from their original scale of measurement, for example, temp = (Fahrenheit-80)/40 so the original values of the predictor were 40, 80 and 120. The data is presented in John (1971) and give an example of a *central composite* design. We compute the covariance of the predictors:

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```
> cov(odor[,-1])
        temp     gas    pack
temp 0.57143 0.00000 0.00000
gas  0.00000 0.57143 0.00000
pack 0.00000 0.00000 0.57143
```

The matrix is diagonal. Even if temp was measured in the original Fahrenheit scale, the matrix would still be diagonal, but the entry in the matrix corresponding to temp would change. Now fit a model, while asking for the correlation of the coefficients:

```
> lmod <- lm(odor ~ temp + gas + pack, odor)
> summary(lmod,cor=T)
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)     15.2        9.3    1.63     0.13
temp           -12.1       12.7   -0.95     0.36
gas            -17.0       12.7   -1.34     0.21
pack           -21.4       12.7   -1.68     0.12

Residual standard error: 36 on 11 degrees of freedom
Multiple R-Squared: 0.334,      Adjusted R-squared: 0.152
F-statistic: 1.84 on 3 and 11 DF,  p-value: 0.199

Correlation of Coefficients:
     (Intercept) temp gas
temp 0.00
gas  0.00        0.00
pack 0.00        0.00 0.00
```

We see that, as expected, the pairwise correlation of all the coefficients is zero. Notice that the SEs for the coefficients are equal due to the balanced design. Now drop one of the variables:

```
> lmod <- lm(odor ~ gas + pack, odor)
> summary(lmod)
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    15.20       9.26    1.64     0.13
gas           -17.00      12.68   -1.34     0.20
pack          -21.37      12.68   -1.69     0.12

Residual standard error: 35.9 on 12 degrees of freedom
Multiple R-Squared: 0.279,      Adjusted R-squared: 0.159
F-statistic: 2.32 on 2 and 12 DF,  p-value: 0.141
```

The coefficients themselves do not change, but the residual SE does change slightly, which causes small changes in the SEs of the coefficients, *t*-statistics and p-values, but nowhere near enough to change our qualitative conclusions.

Exercises

1. The dataset teengamb concerns a study of teenage gambling in Britain. Fit a regression model with the expenditure on gambling as the response and the sex, status, income and verbal score as predictors. Present the output.

   (a) What percentage of variation in the response is explained by these predictors?

   (b) Which observation has the largest (positive) residual? Give the case number.

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(c) Compute the mean and median of the residuals.

(d) Compute the correlation of the residuals with the fitted values.

(e) Compute the correlation of the residuals with the income.

(f) For all other predictors held constant, what would be the difference in predicted expenditure on gambling for a male compared to a female?

2. The dataset uswages is drawn as a sample from the Current Population Survey in 1988. Fit a model with weekly wages as the response and years of education and experience as predictors. Report and give a simple interpretation to the regression coefficient for years of education. Now fit the same model but with logged weekly wages. Give an interpretation to the regression coefficient for years of education. Which interpretation is more natural?
3. In this question, we investigate the relative merits of methods for computing the coefficients. Generate some artificial data by:

```
> x <- 1:20
> y <- x+rnorm(20)
```

Fit a polynomial in x for predicting y. Compute cTβ^ in two ways — by lm() and by using the direct calculation described in the chapter. At what degree of polynomial does the direct calculation method fail? (Note the need for the I() function in fitting the polynomial, that is, lm(y ~ x + l(x^2)) .

4. The dataset prostate comes from a study on 97 men with prostate cancer who were due to receive a radical prostatectomy. Fit a model with lpsa as the response and lcavol as the predictor. Record the residual standard error and the R^2. Now add lweight, svi, lbph, age, lcp, pgg45 and gleason to the model one at a time. For each model record the residual standard error and the R^2. Plot the trends in these two statistics.
5. Using the prostate data, plot lpsa against lcavol. Fit the regressions of lpsa on lcavol and lcavol on lpsa. Display both regression lines on the plot. At what point do the two lines intersect?
6. Thirty samples of cheddar cheese were analyzed for their content of acetic acid, hydrogen sulfide and lactic acid. Each sample was tasted and scored by a panel of judges and the average taste score produced. Use the cheddar data to answer the following:

(a) Fit a regression model with taste as the response and the three chemical contents as predictors. Report the values of the regression coefficients.

(b) Compute the correlation between the fitted values and the response. Square it. Identify where this value appears in the regression output.

(c) Fit the same regression model but without an intercept term. What is the value of R^2 reported in the output? Compute a more reasonable measure of the goodness of fit for this example.

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(d) Compute the regression coefficients from the original fit using the QR decomposition showing your R code.

7. An experiment was conducted to determine the effect of four factors on the resistivity of a semiconductor wafer. The data is found in wafer where each of the four factors is coded as − or + depending on whether the low or the high setting for that factor was used. Fit the linear model resist ~ x1 + x2 + x3 + x4.

(a) Extract the X matrix using the model.matrix function. Examine this to determine how the low and high levels have been coded in the model.

(b) Compute the correlation in the X matrix. Why are there some missing values in the matrix?

(c) What difference in resistance is expected when moving from the low to the high level of x1?

(d) Refit the model without x4 and examine the regression coefficients and standard errors? What stayed the the same as the original fit and what changed?

(e) Explain how the change in the regression coefficients is related to the correlation matrix of X.

8. An experiment was conducted to examine factors that might affect the height of leaf springs in the suspension of trucks. The data may be found in truck. The five factors in the experiment are set to - and * but it will be more convenient for us to use −1 and +1. This can be achieved for the first factor by:

```
truck$B <- sapply(truck$B, function(x) ifelse(x=="-",-1,1))
```

Repeat for the other four factors.

(a) Fit a linear model for the height in terms of the five factors. Report on the value of the regression coefficients.

(b) Fit a linear model using just factors B, C, D and E and report the coefficients. How do these compare to the previous question? Show how we could have anticipated this result by examining the X matrix.

(c) Construct a new predictor called A which is set to B+C+D+E. Fit a linear model with the predictors A, B, C, D, E and 0. Do coefficients for all six predictors appear in the regression summary? Explain.

(d) Extract the model matrix X from the previous model. Attempt to compute $\hat{\beta}$ from $(X^TX)^{-1}X^Ty$. What went wrong and why?

(e) Use the QR decomposition method as seen in Section 2.7 to compute $\hat{\beta}$. Are the results satisfactory?

(f) Use the function qr.coef to correctly compute $\hat{\beta}$.

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